🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 · STANDARD 6.GR.4

Determine Surface Area of Pyramids

Lesson 5-8 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the surface area of a pyramid by adding the base area and the lateral faces.

Language Objective: I can explain my work using the words pyramid, slant height, lateral face, and base.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Pyramid | Slant height | Lateral face | Base | Apex | Lateral area

👆 *Tap any word to see its meaning and a picture.*

1. A solid with a polygon base and triangular faces meeting at one point is a ________.

2. The height of a triangular face of a pyramid, measured along the slanted side, is the ________.

3. A triangular side face of a pyramid is a ________.

4. The polygon on the bottom of a pyramid is the ________.

5. The single top point where the triangular faces of a pyramid meet is the ________.

6 The combined area of all the side faces, not counting the base, is the

____________________ .

Write About the Math

How much will the glass for all 4 triangular sides cost? (The base is open — no glass needed.)?

¿Cuánto costará el vidrio de los 4 lados triangulares? (La base está abierta — no necesita vidrio.)?

Say your answer to a partner first. Then write it, using at least two of these words:

- [ ] **Pyramid** *Pirámide*
- [ ] **Slant height** *Altura inclinada (apotema)*
- [ ] **Lateral face** *Cara lateral*
- [ ] **Base** *Base*
- [ ] **Apex** *Ápice*

Model: *Surface area of a pyramid is the base area plus the areas of all the triangular lateral faces.* — Students explain a square pyramid's surface area = base area + the four triangular lateral faces, and that you add every face together.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Each triangular face has area = ½ × ____ × ____ = ____ ft². Four faces: 4 × ____ = ____ ft². Cost: ____ × $12 = $____.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Pyramid
2. **Notes 2:** Slant height
3. **Notes 3:** Lateral face
4. **Notes 4:** Base
5. **Notes 5:** Apex
6. **Notes 6:** Lateral area
7. **Watch for this mistake:** *A common mistake in Surface Area of Pyramids is finding the four triangular lateral faces correctly (½ × base edge × slant height, times 4) and then stopping — forgetting to add the base area at the end. This leaves the total surface area too small by exactly the base's area. Before you submit, ask: 'Did I add the base area to the four lateral faces, or did I stop after the sides?'*
8. **Try It 1:** C. 81 in² — *The base is a square, so base area = side × side = 9 × 9 = 81 in². You would still add the four triangular faces to get the full surface area.*
9. **Try It 2:**